

Energy-Efficient Beamforming Based on Statistical CSI for Multicast LEO Satellite Communications

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Abstract—Low Earth Orbit (LEO) satellites have recently attracted attention as solutions for enabling global connectivity because of the low propagation latency and low path loss compared to other satellite systems. In the meantime, however, the high orbital speed as well as the low production cost have rendered power consumption minimization as one of the most critical issues for performance optimization in LEO satellite communications. In this paper, we investigate energy-efficient beamforming design for LEO satellites that support multigroup multicast downlink communications. Different from related work that assumes availability of instantaneous channel state information (CSI), our design relies only on the slow-varying statistical CSI (sCSI). The formulated problem aims to maximize energy efficiency while meeting the target data rate requirements. To solve the problem, we first approximate the average rate function by applying a closed-form tight upper bound. We then use the Dinkelbach's algorithm to transform the fractional function for energy efficiency into a concave function. We finally adopt the Convex-Concave Procedure (CCCP) to solve the problem. Numerical results show that the proposed algorithm is able to achieve higher energy efficiency than existing approaches for multicast LEO satellite communications.

I. INTRODUCTION

Compared with the Geostationary Earth Orbit (GEO) satellite or the Medium Earth Orbit (MEO) satellite, Low Earth Orbit (LEO) satellites have recently attracted attention due to the lower propagation latency, lower path loss, higher throughput, and lower production costs. The role of LEO satellites is expected to be multifaceted, including (i) supporting the provision of terrestrial network services in unserved areas (remote areas, aircraft, or ships) as well as in under-served areas (e.g., suburban/rural areas); (ii) improving terrestrial network service reliability through better service continuity; and (iii) achieving terrestrial network scalability by providing efficient multicast/broadcast resources for data transmission [1]. Support for multicast service has in particular caught the limelight due to the substantial increase in demand for content-centric applications such as mobile video streaming [2].

It has been reported that the traffic of video applications has occupied 70% of the total network traffic, and it is expected to be increasing in the future [3]. While satellite communication has developed advanced multicast technology in use

for many years [4], there are still problems to be solved in recent content-centric multicast scenarios, including effective and adaptive clustering of terrestrial users, efficient allocation of energy and radio resources, and optimal design of energy-efficient beamforming for achieving the desired service quality. Notwithstanding the abundance of related work on unicast communications, research on multicast transmission for LEO satellite communication (SatCom) is relatively under-explored.

Ideally, optimal beamforming design should be based on the instantaneous CSI (iCSI) of the channel between the satellite and the user terminal (UT). However, in LEO SatCom scenarios, it is challenging to obtain iCSI due to the high moving speed of LEO satellites and the long propagation delay between a satellite and the UT. A more practical approach [5] investigated in recent research is to use statistical CSI (sCSI) for solving this problem instead. Using sCSI not only reduces the energy and computing resources needed to estimate iCSI, but also improves the robustness of beamforming design in the LEO satellite fast-varying channel. Related work that relies on iCSI for performance optimization in LEO SatCom thus falls short from this perspective.

We note that the source of power that satellites can use mainly comes from the sun. However, LEO satellites are typically blocked by the Earth for about 34% of each orbital time, during which the battery is the main energy supply. Minimizing the energy consumption of a satellite is hence an important issue that needs to be addressed. While related work exists that considers energy efficiency (EE) as the primary optimization metric in beamforming design for multicast communications, their designs have limitations and the achievable EE cannot be maximized as we discuss in Section II.

Inspired by the aforementioned discussion, this paper investigates the problem of optimal beamforming design with the objective of maximizing EE for downlink multigroup multicast LEO SatCom systems. A QoS (quality of service) constraint on the sum rate is imposed to avoid the achieved sum rates for some UTs being too low due to disadvantaged EE. We consider the use of only sCSI in the system design to lighten the burden of collecting iCSI. The problem thus formulated on energy-efficient beamforming design is difficult to solve for the following reasons: (i) the objective function has only an

This work was supported in part by funds from the National Science and Technology Council and National Taiwan University under Grant NSTC-113-2221-E-002-139 and Grant NSTC-113-2218-E-002-038.

average rate function due to the use of sCSI, and (ii) the EE maximization problem is a fractional programming problem with a nonconvex constraint. To tackle these problems, we first use a closed-form tight approximation for the average rate function. We then use the Dinkelbach's algorithm to transform the fractional function into a concave function. Finally, we adopt the Convex-Concave Procedure (CCCP) to solve the formulated problem. Numerical results presented in Section V show that the achieved EE in the proposed approach is stable and better than that presented in related work.

II. RELATED WORK

While the beamforming technology for terrestrial networks has been widely studied, owing to the high mobility and long transmission distance characteristics in LEO SatCom, beamforming for SatCom is rather different [5]. Many research endeavors for LEO SatCom have focused on unicast transmission, yet research for multicast transmission is relatively under-explored. This paper aims to bridge the gap by focusing on beamforming design for LEO satellite multicast transmission.

Related work has investigated the SatCom multicast beamforming design by maximizing the sum rate [2], [6], minimizing transmit power [7], or maximizing EE [8]. To obtain the system max-min fair capacity, [2] jointly designs downlink beamforming and subchannel assignment. The authors adopt a many-to-many matching model to address a non-convex problem that is guaranteed to converge within finite iteration steps. [6] proposes a joint grouping and beamforming design method to maximize the sum rate. [7] designs a power-domain non-orthogonal multiple access (NOMA) beamforming to minimize the total transmit power of the integrated satellite-terrestrial Internet-of-Things (IoT) network. The proposed algorithm ensures high spectrum efficiency and robustness. [8] maximizes EE by considering rate QoS constraint. However, all the above works assume perfect iCSI that is impractical in the LEO SatCom network as we have mentioned.

There does exist related work [9], [10] that has considered sCSI and EE in the multicast SatCom scenario as we have targeted. However, their design cannot achieve the maximum EE value. For example, [9] aims to minimize beamforming power while satisfying the required EE lower bound. While minimizing beamforming power can potentially increase EE, these two objectives are not equivalent. Other power drains such as inherent circuit power may eventually outweigh the beamforming power and make minimizing beamforming power ineffective towards EE maximization. The authors in [10] aim to maximize EE while avoiding Power-Consumption Outage (PCO). Hence, the formulated problem imposes a sum rate upper bound that in turns limits the maximum EE that can be achieved. Different from these endeavors, our work aims to maximize EE while meeting the requirement in sum rate for LEO SatCom as we explain in the following.

III. SYSTEM MODELS AND PROBLEM FORMULATION

We consider a full-frequency reuse downlink LEO satellite system that supports multigroup multicast communications as

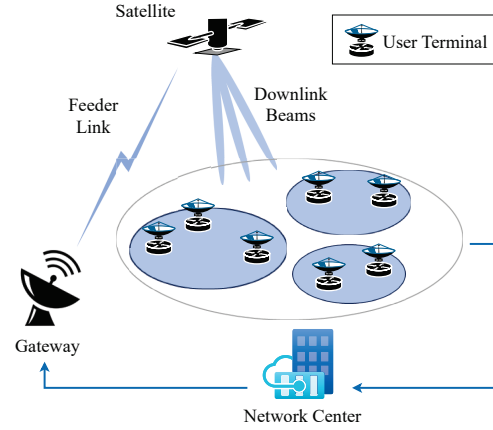


Fig. 1: Multicast multigroup LEO satellite communications.

illustrated in Fig. 1. When UTs receive the signal from the satellite, they estimate downlink CSI and transmit it to the network center. The network center calculates the beamforming weight for the satellite based on the CSI and then transmits it to the gateway that in turn conveys the weight to satellites through feeder links. Fig. 2 further shows the overall loop of transmission process. Although this entire process may cause additional delay compared to UTs directly transmitting the CSI to satellites and satellites calculating beamforming weight on-board, it can lower the load of satellites for computing the beamforming weight. Due to the transmitting distance of satellites being too long, iCSI could be outdated when received. Therefore, a better way is to leverage outdated iCSI to estimate the statistical properties of the channel and then use sCSI for beamforming design.

A. Channel Model

We assume that each LEO satellite is equipped with a uniform planar array (UPA) and each UT is equipped with one antenna. The UPA consists of $N = N_x \times N_y$ antenna, where N_x and N_y are the number of antennae on the x and y axes, respectively. The channel model for the k th UT at instant t and frequency f thus can be described as follows:

$$\mathbf{h}_k(t, f) = h_k(t, f) \cdot \mathbf{v}_k, \quad (1)$$

where $\mathbf{v}_k \in \mathbb{C}^{N \times 1}$ is the UPA array response vector. The UPA array response vector $\mathbf{v}_k$ associated with the k th UT can be described as follows:

$$\mathbf{v}_k = \mathbf{v}_x(\theta_k^x) \otimes \mathbf{v}_y(\theta_k^y), \quad (2)$$

where $\theta_k^x = \sin\phi_k^y \cos\phi_k^x$, $\theta_k^y = \cos\phi_k^y$, and

$$\mathbf{v}_d(\theta_k^d) = \frac{1}{\sqrt{N_d}} \left[1 e^{-j\pi\theta_k^d} \dots e^{-j\pi(N_d-1)\theta_k^d} \right], \quad (3)$$

with $d \in \{x, y\}$. ϕ_k^x and ϕ_k^y represent the azimuth angle and elevation angle related to the k th UT respectively.

For modeling the channel gain $h_k(t, f)$ in (1), related work has adopted a multipath channel model between the LEO satellite and the UT, with the assumption that the Doppler shift caused by the motion of the LEO satellite and the minimum propagation delays between the UT and satellite can be compensated at the receiver [5]. In this way, $h_k(t, f)$ in (1) is formulated as follows:

$$h_k(t, f) = \sum_{p=1}^{P_k} \alpha_{k,p} \exp\{j2\pi[t f_{k,p} - f \tau_{k,p}]\}, \quad (4)$$

where P_k denotes the number of propagation paths, while $\alpha_{k,p}$, $f_{k,p}$, and $\tau_{k,p}$ are the complex channel gain, Doppler shift, and propagation delay of the k th UT in the p th path respectively.

It is worth noting that since $h_k(t, f)$ is difficult to obtain instantaneously as iCSI, in this paper we rely only on sCSI to capture the statistical properties of the channel instead. In particular, the Rician channel model is used popularly in the literature to model LoS propagation in LEO SatCom systems. Hence, we model the channel gain $h_k(t, f)$ based on the Rician fading distribution with the Rician factor κ_k and power $E\{|h_k(t, f)|^2\} = \gamma_k$. The real and imaginary parts of $h_k(t, f)$ are independently and identically real-valued Gaussian distributed with mean $\sqrt{\frac{\kappa_k \gamma_k}{2(\kappa_k + 1)}}$ and variance $\frac{\gamma_k}{2(\kappa_k + 1)}$ respectively. In other words, we assume knowledge of

$$E\{||\mathbf{h}_k(t, f)||^2\} = E\{||h_k(t, f) \cdot \mathbf{v}_k||^2\} = ||\mathbf{v}_k||^2 \gamma_k \quad (5)$$

instead of $\mathbf{h}_k(t, f)$ itself in this paper. For sake of notation simplicity, we use $\mathbf{h}_k$ to refer to $\mathbf{h}_k(t, f)$ in the following.

B. Transmission Model

The LEO satellite has N antenna and L (with $L \leq N$) RF links (beams) covering L multicast groups and K active UTs, where $K \geq L$. The index set of all UTs and multicast groups are denoted as $\mathcal{K} = \{1, \dots, k, \dots, K\}$ and $\mathcal{L} = \{1, \dots, l, \dots, L\}$, respectively. Note that which UT k belongs to which group l is determined according to the grouping result. Let U_l represent the set of UTs in the l th multicast group and $|U_l|$ represent the number of UT in the l th multicast group. It should be noted that each UT belongs to only one multicast group, which can be denoted as $U_l \cup U_j = \emptyset, \forall l, j \in \{1, \dots, L\}, l \neq j$. In this paper, to keep the scope focused, we adopt the hierarchical clustering algorithm [11] to cluster the UTs, where UTs are grouped together if their channel information is similar. We assume the grouping process is done beforehand in this paper.

We use Fig. 2 as an example to describe the transmission model with two groups and four UTs. UT 1 and UT 4 belong to group 1, while UT 2 and UT 3 belong to group 2. The received signal $y_{k,l}$ of the k th UT in l th group can be expressed as:

$$y_{k,l} = \mathbf{h}_k^H \mathbf{f}_l s_l + \sum_{j=1, j \neq l}^L \mathbf{h}_k^H \mathbf{f}_j s_j + n_k, \quad k \in U_l, l \in \mathcal{L}, \quad (6)$$

where the first term in (6) represents the desirable received signal of the k th UT in the l th multicast group, the second term represents the interference of another multicast group, and the

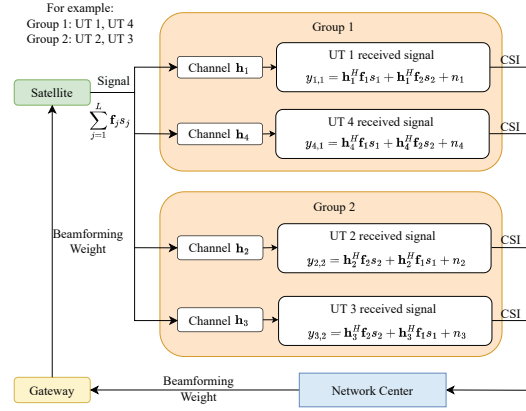


Fig. 2: Illustration of transmission scenario.

third term represents the noise; $\mathbf{f}_l \in \mathbb{C}^{N \times 1}$ is the beamforming weight vector of the l th multicast group, $\mathbf{h}_k \in \mathbb{C}^{N \times 1}$ represents the channel vector, s_l represents the signal of the multicast group U_l with the unit power constraint $E\{|s_l|^2\} = 1$, and $n_k \sim \mathcal{CN}(0, N_0)$ is the additive white Gaussian noise of the k th UT.

C. System Rate

According to the received signal model in (6), the SINR of UT k in the l th multicast group can be defined as

$$\text{SINR}_{k,l} = \frac{|\mathbf{f}_l^H \mathbf{h}_k|^2}{\sum_{j \neq l} |\mathbf{f}_j^H \mathbf{h}_k|^2 + N_0}, \quad \forall k \in U_l, j, l \in \mathcal{L}. \quad (7)$$

For profiling performance of long-term transmission, we use the average data rate of the k th UT in the l th group as follows:

$$R_{k,l} = \mathbb{E}\{\log(1 + \text{SINR}_{k,l})\}. \quad (8)$$

Note that using $\text{SINR}_{k,l}$ directly for optimization requires knowing iCSI $\mathbf{h}_k$. To avoid this pitfall, we approximate $R_{k,l}$ as $\overline{R_{k,l}}$ based on the approximate method in [12] as follows:

$$\begin{aligned} \overline{R_{k,l}} &\triangleq \log \left(1 + \frac{\mathbb{E}\{|\mathbf{f}_l^H \mathbf{h}_k|^2\}}{\mathbb{E}\{\sum_{j \neq l} |\mathbf{f}_j^H \mathbf{h}_k|^2\} + N_0} \right) \\ &= \log \left(1 + \frac{|\mathbf{f}_l^H \mathbf{v}_k|^2 \gamma_k}{\sum_{j \neq l} |\mathbf{f}_j^H \mathbf{v}_k|^2 \gamma_k + N_0} \right), \\ &= \log(1 + r_{k,l}), \end{aligned} \quad (9)$$

where the second equality is obtained using (5) and

$$r_{k,l} = \frac{|\mathbf{f}_l^H \mathbf{v}_k|^2 \gamma_k}{\sum_{j \neq l} |\mathbf{f}_j^H \mathbf{v}_k|^2 \gamma_k + N_0}, \quad \forall k, l. \quad (10)$$

With the approximation, we only need to know sCSI γ_k rather than iCSI $\mathbf{h}_k$ for performance optimization.

In the multicast transmission scenario, for multicast group l , the achievable rate is determined by the UT with the minimum rate in this group as follows:

$$R_l = \min_{k \in U_l} \overline{R_{k,l}}. \quad (11)$$

D. Energy Efficiency

The EE of the whole system can be defined as

$$EE = \frac{B_w \sum_{l=1}^L |U_l| R_l}{P_{\text{total}}} = \frac{B_w \sum_{l=1}^L |U_l| R_l}{\sum_{l=1}^L \|\mathbf{f}_l\|^2 + P_0}, \quad (12)$$

where B_w represents the bandwidth, $|U_l|$ represents the number of UT in group l and $P_{\text{total}} = \sum_{l=1}^L \|\mathbf{f}_l\|^2 + P_0$ represents the total power consumption, which includes the transmission beamforming power of the LEO satellite $\sum_{l=1}^L \|\mathbf{f}_l\|^2$ and the inherent power consumption of the communication system P_0 . Here, we assume that P_0 is a constant to keep the scope of this paper focused.

E. Problem Description

Under the transmission power and QoS constraints, our goal is to maximize the EE by designing beamformers for each multicast group. The problem can be formulated as follows:

$$\begin{aligned} & \text{maximize} \quad \frac{B_w \sum_{l=1}^L |U_l| R_l}{\sum_{l=1}^L \|\mathbf{f}_l\|^2 + P_0} \\ & \text{subject to} \quad \text{C1 : } R_l \geq R_0, \forall l, \\ & \quad \quad \quad \text{C2 : } \sum_{l=1}^L \|\mathbf{f}_l\|^2 \leq P, \end{aligned} \quad (13)$$

where C1 is the QoS constraint with R_0 being the minimum transmission rate requirement, and C2 is the beamforming power constraint with P being the maximum power.

IV. SOLVING THE OPTIMIZATION PROBLEM

Since the problem (13) is difficult to solve, we refer to [6] by introducing a slack variable $\tilde{r}_l$ for each group and transform the problem to the following:

$$\begin{aligned} & \text{maximize} \quad \frac{B_w \sum_{l=1}^L |U_l| \log(1 + \tilde{r}_l)}{\sum_{l=1}^L \|\mathbf{f}_l\|^2 + P_0} \\ & \text{subject to} \quad \text{C1 : } r_{k,l} \geq \tilde{r}_l, \forall k, \forall l, \\ & \quad \quad \quad \text{C2 : } \tilde{r}_l \geq r_0, \forall l, \\ & \quad \quad \quad \text{C3 : } \sum_{l=1}^L \|\mathbf{f}_l\|^2 \leq P, \end{aligned} \quad (14)$$

where $r_0 = 2^{R_0} - 1$. This step removes the minimization function in (11). Denoting $\mathbf{F} \in \mathbb{C}^{N \times L} = [\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_L]$, we rewrite C1 to be a difference of two convex functions as follow:

$$\begin{aligned} & \text{maximize} \quad \frac{B_w \sum_{l=1}^L |U_l| \log(1 + \tilde{r}_l)}{\sum_{l=1}^L \|\mathbf{f}_l\|^2 + P_0} \\ & \text{subject to} \quad \text{C1 : } W_k(\mathbf{F}) - G_k(\mathbf{F}, \tilde{r}_l) \leq 0, \forall k, \forall l, \\ & \quad \quad \quad \text{C2 : } \tilde{r}_l \geq r_0, \forall l, \\ & \quad \quad \quad \text{C3 : } \sum_{l=1}^L \|\mathbf{f}_l\|^2 \leq P, \end{aligned} \quad (15)$$

where $W_k(\mathbf{F}) = \sum_{j \neq l} |\mathbf{f}_j^H \mathbf{v}_k|^2 \gamma_k + N_0$ and $G_k(\mathbf{F}, \tilde{r}_l) = (\sum_{j=1}^L |\mathbf{f}_j^H \mathbf{v}_k|^2 \gamma_k + N_0) / (1 + \tilde{r}_l)$.

The objective function of the above problem is a concave-convex fractional function. We use the Dinkelbach's algorithm [13] to tackle the concave-convex fractional function. Our problem can be transformed to maximize

$$(B_w \sum_{l=1}^L |U_l| \log(1 + \tilde{r}_l)) - \lambda (\sum_{l=1}^L \|\mathbf{f}_l\|^2 + P_0),$$

which is a concave function. We then rewrite our problem as follows:

$$\begin{aligned} & \text{maximize} \quad (B_w \sum_{l=1}^L |U_l| \log(1 + \tilde{r}_l)) - \lambda (\sum_{l=1}^L \|\mathbf{f}_l\|^2 + P_0) \\ & \text{subject to} \quad \text{C1 : } W_k(\mathbf{F}) - G_k(\mathbf{F}, \tilde{r}_l) \leq 0, \forall k, \forall l, \\ & \quad \quad \quad \text{C2 : } \tilde{r}_l \geq r_0, \forall l, \\ & \quad \quad \quad \text{C3 : } \sum_{l=1}^L \|\mathbf{f}_l\|^2 \leq P. \end{aligned} \quad (16)$$

We find that (16) is a difference of convex programming, and the Convex-Concave Procedure [14] can be used to solve it. We linearize the concave part $G_k(\mathbf{F}, \tilde{r}_l)$ in C1 in (14) using the first-order Taylor approximation as follows:

$$\begin{aligned} \hat{G}_k(\mathbf{F}^i, \tilde{r}_l^i) &= G_k(\mathbf{F}^{i-1}, \tilde{r}_l^{i-1}) \\ &+ \nabla^T G_k(\mathbf{F}^{i-1}, \tilde{r}_l^{i-1}) \begin{bmatrix} \mathbf{f}_L^i - \mathbf{f}_L^{i-1} \\ \vdots \\ \mathbf{f}_1^i - \mathbf{f}_1^{i-1} \\ \tilde{r}_l^i - \tilde{r}_l^{i-1} \end{bmatrix}, \end{aligned} \quad (17)$$

where i represents the i th iteration in CCCP and

$$\nabla G_k(\mathbf{F}^{i-1}, \tilde{r}_l^{i-1}) = \begin{bmatrix} \frac{2\mathbf{v}_k^H \mathbf{f}_L^{i-1} \mathbf{v}_k^* \gamma_k}{1 + \tilde{r}_l^{i-1}} \\ \vdots \\ \frac{2\mathbf{v}_k^H \mathbf{f}_1^{i-1} \mathbf{v}_k^* \gamma_k}{1 + \tilde{r}_l^{i-1}} \\ -\frac{\sum_{j=1}^L |\mathbf{v}_k^H \mathbf{f}_j^{i-1}|^2 \gamma_k + N_0}{(1 + \tilde{r}_l^{i-1})^2} \end{bmatrix}. \quad (18)$$

Finally, the problem to solve for i th CCCP iteration can be rewritten as follows:

$$\begin{aligned} & \text{maximize} \quad (B_w \sum_{l=1}^L |U_l| \log(1 + \tilde{r}_l^i)) - \lambda (\sum_{l=1}^L \|\mathbf{f}_l^i\|^2 + P_0) \\ & \text{subject to} \quad \text{C1 : } W_k(\mathbf{F}^i) - \hat{G}_k(\mathbf{F}^i, \tilde{r}_l^i) \leq 0, \forall k, \forall l, \\ & \quad \quad \quad \text{C2 : } \tilde{r}_l^i \geq r_0, \forall l, \\ & \quad \quad \quad \text{C3 : } \sum_{l=1}^L \|\mathbf{f}_l^i\|^2 \leq P, \end{aligned} \quad (19)$$

where $\hat{G}_k(\mathbf{F}^i, \tilde{r}_l^i)$ is in (17).

We summarize the full algorithm for solving the formulated problem in Algorithm 1. The outer **while** loop is the Dinkelbach's algorithm, and the inner **while** loop (line 5 to line 9)

is the CCCP algorithm. Variables n and i are the indices of the outer and inner while loops. Note that the initial value of the CCCP problem could significantly affect the converged value. The initial beamforming weight $\mathbf{F}^0$ can be randomly generated to satisfy constraint C3 in (15). The initial slack variable $\tilde{r}_l^0 = \min_{k \in U_l} r_{k,l}^0$, where $r_{k,l}^0$ is obtained using (10) and the initial beamforming weight $\mathbf{F}^0$.

Algorithm 1 Energy-Efficient Beamforming Algorithm

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1:  $\epsilon > 0, n = 0, \lambda_0 = 0$ 
2: while  $A(\lambda_n) > \epsilon$  do
3:   given an initial feasible point  $(\mathbf{F}^0, \tilde{r}_l^0)$ 
4:    $i = 0$ 
5:   while Problem (19) convergence do
6:     Obtain  $\hat{G}_k(\mathbf{F}^i, \tilde{r}_l^i)$  by (17)
7:     Solve (19) to obtain optimal values  $\{\mathbf{f}_l^i, \tilde{r}_l^i\}_{l=1}^L$ 
8:     Update iteration:  $i = i + 1$ 
9:   end while
10:   $A(\lambda_n) = \left( B_w \sum_{l=1}^L |U_l| \log(1 + \tilde{r}_l^n) \right) -$ 
     $\lambda_n \left( \sum_{l=1}^L \|\mathbf{f}_l^n\|^2 + P_0 \right)$ 
11:   $\lambda_{n+1} = \frac{B_w \sum_{l=1}^L |U_l| \log(1 + \tilde{r}_l^n)}{\sum_{l=1}^L \|\mathbf{f}_l^n\|^2 + P_0}$ 
12:   $n = n + 1$ 
13: end while
    
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V. EVALUATION RESULTS

In this section, we evaluate the performance of the proposed algorithm. We randomly generate 21 UTs in a 200×200 km square space and cluster them into 5 groups based on the channel information. The satellite is located at the center and has a 1000 km height. We assume that all UTs have the same Rician factor, i.e., $\kappa_k = \kappa = 18$ dB, $\forall k$. The channel power can be given by $\gamma_k = NG_t G_r \left(\frac{v_c}{4\pi f_c d_k} \right)^2$, $\forall k$, where G_t and G_r are the antenna gains of the transmitter and the receiver respectively, f_c , v_c , d_k are the carrier frequency, speed of light, and the distance between the k th UT and the LEO satellite respectively. The noise power N_0 is characterized by $N_0 = k_B B_w T_n$, where k_B and T_n denote the Boltzmann constant and the noise temperature, respectively. All simulation parameters listed in Table I are from references [5] and [6].

We compare our performance with two state-of-the-art methods in multicast SatCom, namely, high throughput satellite communication system (HTSCS) [10] and robust energy-efficient beamforming algorithm (REEB) [9]. HTSCS maximizes EE by considering a PCO constraint and beamforming power constraint. REEB minimizes beamforming power by considering a lower EE bound. Our formulation, on the other hand, maximizes EE while meeting the sum rate lower bound. We vary the SNR to analyze the performance of different methods. The SNR is calculated by the signal power $E\{|s|^2\}$ multiplied by free space path loss γ divided by the noise power N_0 . We adjust the noise power and put it into (7) to observe the performance of different SNRs.

Parameter	Value
Number of satellite antenna	4×4
System bandwidth	$B_w = 50\text{MHz}$
Carrier frequency	$f_c = 2\text{GHz}$
Antenna gain of satellite	$G_t = 36.7\text{dBi}$
Antenna gain of UT	$G_r = 41.7\text{dBi}$
QoS constraint	$r_0 = 0.12$
Inherent power	$P_0 = 1.895$ Watts
Boltzmann constant	$k_B = 1.38 \times 10^{-23} \text{J} \cdot \text{K}^{-1}$
Noise temperature	$T_n = 300$ K

TABLE I: Simulation Parameters

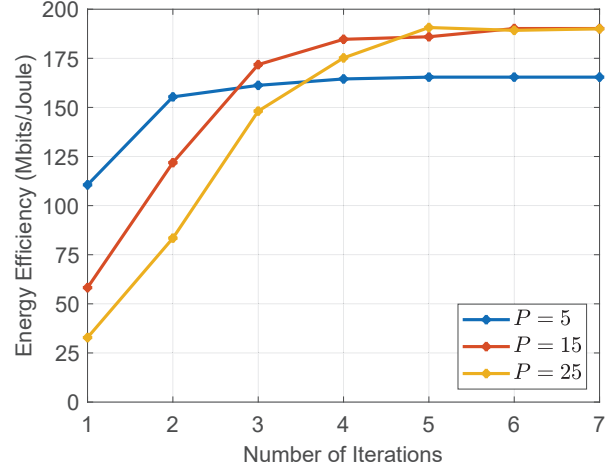


Fig. 3: Convergence performance versus the number of iterations.

Firstly, we show the convergence performance of our algorithm for different power constraints in Fig. 3. Our algorithm has a lower complexity since it can converge in about 7 iterations of the Dinkelbach's while loop. In the first iteration, $\lambda = 0$, so the algorithm tries to maximize the first term in the objective function while neglecting the power term. This causes the EE to be low. Since the initial beamforming is generated based on the maximum beamforming power constraint, the performance with a higher power constraint is worse. However, the EE increases as the iteration continues, and the EE with a higher power constraint becomes higher than the others when the algorithm converges. The reason is that the solutions of the algorithm with a higher power constraint cover the solution range of smaller power constraints, and hence the algorithm with a higher power constraint can find a better solution than that with a lower power constraint.

In Fig. 4, we compare the EE performance of the proposed method with HTSCS and REEB in different SNRs. The maximum beamforming power is set as 25 watts in our algorithm and HTSCS. We can find that our method stands out in the EE performance. Performance of REEB in EE mostly remains around 40, because its algorithm almost stops when the minimum value is met. While EE increases slightly as the

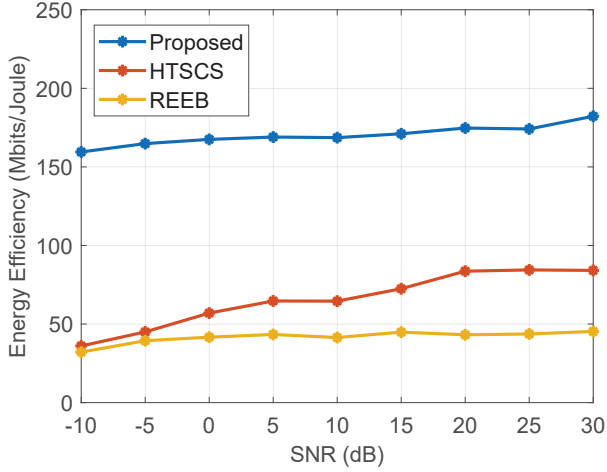


Fig. 4: Comparison of EE performance versus the different SNRs.

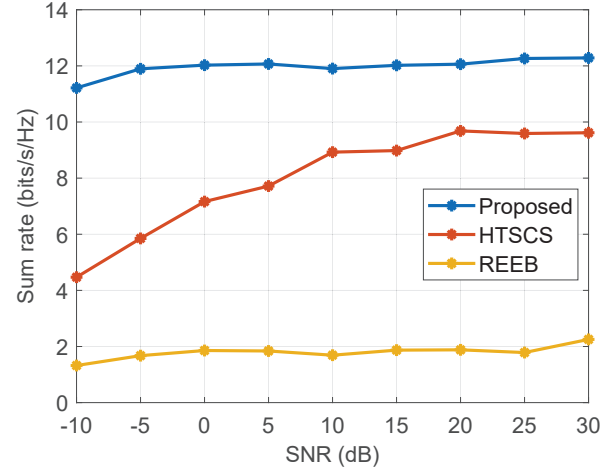


Fig. 5: Comparison of rate performance versus different SNRs.

beamforming power becomes smaller, EE cannot be further improved when the beamforming power is much less than the inherent power. Although the aim of HTSCS is to maximize EE similar to ours, its PCO constraint limits its achievable EE performance.

In Fig. 5, we conduct the simulations in different SNRs to observe the sum rate performance. The maximum beamforming power is set as 25 watts in our algorithm and HTSCS again. We can observe that the performance of the proposed algorithm is higher than HTSCS and REEB. REEB achieves a rate that remains around 2 since its algorithm does not have a rate QoS constraint. Although HTSCS aims to maximize EE where the sum rate can also increase, it cannot achieve a higher rate because of its PCO constraint. The QoS constraint in our beamforming allows the system to perform satisfactorily under bad channel conditions.

VI. CONCLUSION

In this paper, we have investigated the EE maximization beamforming design in the LEO SatCom multigroup multicast scenario based on statistical CSI (sCSI). To solve the formulated problem based on sCSI, a tight upper bound is first used to approximate the average rate function. Secondly, the Dinkelbach's algorithm is adopted to transform a fractional function into a difference function of convex and concave functions. Lastly, a CCCP is used to tackle the difference function between convex and concave functions. Numerical results have demonstrated the EE performance of the proposed algorithm is better than existing approaches.

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